

§9.4 Two-factor analysis of variance

eg. teaching method $\rightarrow$ both influence the quality of
size of class $\rightarrow$ teaching

Data 6 options for factor 2

a options for factor 1

b options for factor 2

X_{ij}

$\bar{X}_{.j}$ (column average)

$\bar{X}_{i.}^2$ (row average)

Assumptions:

- We consider $\boxed{1}$ observation per cell

$$x_{ij} \sim N(\mu_{ij}, \sigma^2)$$

- Additional assumption:

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

$$\begin{aligned} \sum_{i=1}^a \alpha_i &= 0 \\ \sum_{j=1}^b \beta_j &= 0 \end{aligned} \quad \left. \vphantom{\sum_{i=1}^a \alpha_i} \right\} \text{normalization} \quad (\text{see p. 446 Remark})$$

$$SS(TO) = \sum_{i=1}^a \sum_{j=1}^b (x_{ij} - \overset{\text{grand mean}}{\bar{x}})^2 =$$

$$= \sum_{i=1}^a \sum_{j=1}^b [(x_{ij} - \bar{x}_i - \bar{x}_j + \bar{x})^2 + (\bar{x}_i^2 - \bar{x}) + (\bar{x}_j - \bar{x})]^2 =$$

$$= \sum_i^a \sum_j^b (x_{ij} - \bar{x}_i - \bar{x}_j + \bar{x})^2 = \sum_i^a \sum_j^b (\bar{x}_i - \bar{x})^2 + \sum_i^a \sum_j^b (\bar{x}_j - \bar{x})^2$$

$$= \sum_{i=1}^a \sum_{j=1}^b (x_{ij} - \bar{x}_i - \bar{x}_j + \bar{x})^2 + \sum_{i=1}^a (\bar{x}_i - \bar{x})^2 + \sum_{j=1}^b (\bar{x}_j - \bar{x})^2$$

$$SS(E)$$
$$S S'(A)$$
$$SS(B)$$

"row difference"

"column difference"

If μ_A, μ_B are true, $SS(A)$ and $SS(B)$ are χ^2 difference.

$K_A: \alpha_1 = \alpha_2 = \dots = \alpha_n = 0$
"no row effects"

$H_B: \beta_1 = \beta_2 = \dots = \beta_6 = 0$
no column effects

→ $SS(E)$ distribution is "measuring stick", it independes on μ_A, μ_B

Distributions?

$$\frac{SS(TO)}{\sigma^2} = \chi^2(ab-1) \quad (\text{given } \mu_A, \mu_B)$$

$$\frac{SS(A)}{\sigma^2} \sim \chi^2(a-1)$$

$$\frac{SS(B)}{\sigma^2} \sim \chi^2(b-1)$$

$$\frac{SS(E)}{\sigma^2} \sim \chi^2(ab-1 - (a-1) - (b-1)) = \chi^2((a-1)(b-1))$$

Analogously to the ~~previous~~ one-factor,

$$\frac{SS(A)}{a-1} > \sigma^2 \quad \text{if } \mu_A \text{ is not true}$$

$$F_A = \frac{SS(A)/(a-1)}{SS(E)/(a-1)(b-1)} \sim F_{\alpha}(a-1, (a-1)(b-1))$$

$$F_B = \frac{SS(B)/(b-1)}{SS(E)/(a-1)(b-1)} \sim F_{\alpha}(b-1, (a-1)(b-1))$$

"Too large" - reject

Rem!

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

	$\bar{+3}$	$\bar{+4}$	$\bar{+1}$	car type
fuel type	6	3	7	8
	10	7	11	12
	8	5	9	10

$$\begin{aligned} \alpha_1 &= 0 \\ \alpha_2 &= 4 \\ \alpha_3 &= 2 \end{aligned}$$

biggest β_j produces "best" column, α_i produces the "best" row

This does not have to be the case → interactions matter (specific car - specific fuel)

ANOVA table

Sum of Squares	degrees of freedom	MS	F
SS(A)	$a-1$	$SS(A)/(a-1)$	$MS(A)/MS(E)$
SS(B)	$b-1$	$SS(B)/(b-1)$	$MS(B)/MS(E)$
SS(E)	$(a-1)(b-1)$	$SS(E)/(a-1)(b-1)$	
SS(TO)	$ab-1$		

(Ex)

	Gasoline type			
Car	1	2	3	4
1	26	28	31	31
2	24	25	28	27
3	25	25	28	26

"best" car and gasoline

Remark 1

Assumption

$$\mu_{ij} = \mu + \alpha_i + \beta_j$$

What does it mean?

- either all rows and cols fluctuate around same value $\alpha_i's = \beta_j's = 0$

- or $\alpha_1 > \alpha_2$, then row 1 is bigger than row 2 ("car 1 is more efficient in general")

- $\beta_i > \beta_j$ gasoline i is better than gasoline j

What is some car and gasoline are incompatible?

Say, 31 $\rightarrow$ 15 in (1,4)

"Model with interactions"

(car-gas interaction matters)

$$\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$$

assume

$$\begin{cases} \sum_{i=1}^a \gamma_{ij} = 0 \\ \sum_{j=1}^b \gamma_{ij} = 0 \end{cases}$$

Rem 2 can be k observations in each cell $x_{ijk}, i=1, \dots, c$

model with interactions

Hypothesis $H_A: \alpha_i = 0 \quad i=1, \dots, a$

$H_B: \beta_j = 0 \quad j=1, \dots, b$

$H_{AB}: \gamma_{ij} = 0 \quad i, j \text{ 'no interactions'}$

$$SS(TO) = SS(A) + SS(B) + SS(AB) + SS(E)$$

$$SS(A) = bc \sum_{i=1}^a (\bar{X}_i^{2,3} - \bar{X})^2 \sim \sigma^2 x^2(a-1)$$

$$SS(B) = ac \sum_{j=1}^b (\bar{X}_j^{1,3} - \bar{X})^2 \sim \sigma^2 x^2(b-1)$$

$$SS(AB) = c \sum_{i=1}^a \sum_{j=1}^b (\bar{X}_{ij}^3 - \bar{X}_i^{2,3} - \bar{X}_j^{1,3} + \bar{X})^2 \sim \sigma^2 x^2((a-1)(b-1))$$

$$SS(E) = \sum \sum \sum (X_{ijk} - \bar{X}_{ij}^3)^2 \sim \sigma^2 x^2(ab(c-1))$$

$$SS(TO) = \sum \sum \sum (X_{ijk} - \bar{X})^2 \sim x^2(abc-1)$$

Notations $\bar{X} = \frac{1}{abc} \sum_{i,j,k} X_{ijk}$

$$\bar{X}_j^{1,3} = \frac{1}{ac} \sum_{i,k} X_{ijk}$$

$$\bar{X}_{ij}^3 = \frac{1}{c} \sum_{k=1}^c X_{ijk}$$

To test AB,

$$F_{AB} = \frac{MS(AB)}{MS(E)} = \frac{SS(AB)}{SS(E)} \frac{(a-1)(b-1)}{ab(c-1)}$$

$$\sim F((a-1)(b-1), ab(c-1))$$

P. 452 two-way ANOVA table, c observations per cell
Ex. 9.4-2
Problem 9.4-9

Example car + gasoline 9.4-2

$H_A: \mu_1 = \mu_2 = \mu_3 = 0$

1% significance level

$$SS(A) = b \sum_{i=1}^a (\bar{X}_i^2 - \bar{X})^2$$

$$\bar{X} = \frac{1}{12} (26 + 28 + 31 + 31 + 24 + \dots) = 27$$

$$\bar{X}_1^2 = \frac{1}{4} (26 + 28 + 31 + 31) = 29$$

$$\bar{X}_2^2 = 26$$

$$\bar{X}_3^2 = 26$$

$$\bar{X}_4^2 = \frac{1}{3} (26 + 24 + 25) = 25$$

:

$$SS(A) = 4 ((29 - 27)^2 + (26 - 27)^2 + (26 - 27)^2) = 4 \cdot 6 \cdot \textcircled{24}$$

$$SS(E) = (26 - 29 - 25 + 27)^2 + \dots = 4$$

$$a - 1 = 2$$

$$b - 1 = 3$$

$$F = \frac{24/2}{4/6}$$

$$= 18 > F_{0.01}(2, 6) = 10.92$$

reject H_0 (cars are not the same)

See textbook on H_B